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Editor-in-Chief

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Dear Editor-in-Chief and Associate Editors,

We are pleased to submit our original systematic survey manuscript titled:

”Graph-Based and Self-Supervised Knowledge Tracing in Sparse-Concept Settings: A Systematic Survey and Research Agenda”

for consideration for publication as a Survey Paper in *IEEE Transactions on Knowledge and Data Engineering (TKDE)*.

Background and Motivation: Knowledge Tracing (KT) is a fundamental problem in educational data mining and artificial intelligence in education (AIED). While deep learning sequence models have improved accuracy, they suffer from representation collapse under interaction data sparsity and fail to model zero-exposure concept cold-start. In response, Graph Neural Networks (GNNs) and Self-Supervised Learning (SSL) have expanded rapidly over recent years.

Key Contributions of Our Manuscript:

1. **Transparent PRISMA 2020 Protocol:** We present a systematic review across 7 electronic databases (Scopus, Web of Science, ACM DL, IEEE Xplore, ScienceDirect, SpringerLink, arXiv), auditing 2,017 candidate records to establish an integrated core corpus of 36 primary core studies published between 2015 and 2026.
2. **Unified 2D Taxonomy:** We establish a comprehensive two-dimensional taxonomy framework classifying Graph-KT architectures (GNN encoders, provenance, dynamic fusion) and SSL-KT objectives (multi-view contrast, graph structural contrast, generative masked graph auto-encoders).
3. **Critical Evaluation Protocol Audit:** We uncover a critical methodological flaw in the literature: while 94.4% of studies cite data sparsity, only 5.6% evaluate strict zero-exposure concept cold-start protocols.
4. **Probability Calibration Audit:** We conduct the first quality audit revealing that 0 out of 36 core studies evaluate Expected Calibration Error (ECE).
5. **Evidence-Derived Research Agenda:** We outline 6 concrete future research pillars, highlighting LLM-GNN multimodal fusion, non-Euclidean hyperbolic state manifolds, and calibrated trustworthy KT.

Publication & Ethical Statements: We confirm that this manuscript is original, has not been published previously, and is not under consideration for publication elsewhere. All authors have read and approved the submitted version of the manuscript.

Thank you for considering our work for publication in IEEE TKDE. We look forward to receiving feedback from you and the reviewers.

Sincerely yours,

Khanh-Trinh Nguyen (First Author)

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On behalf of all co-authors:

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